The Chladni Plate

A brief introduction to some continuum mechanics and the chladni plate experiment

Vivek Kalita

Department of Physics, IISER Mohali

10 November 2026

Contents

Some Basics **1**

1.1 Strain 1

1.2 Strain 4

1.3 Newton's laws and the Navier–Cauchy equation 5

The Chladni Plate **7**

1 Some Basics

1.1 Strain

Consider a continuous body occupying a region of three-dimensional space. Every material point in the body is specified by its position vector

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

A deformation of the body can be described mathematically as a mapping in which every material point is assigned a new position. We write this mapping as

$$\mathbf{r} \longmapsto \mathbf{r} + \mathbf{u}(\mathbf{r})$$

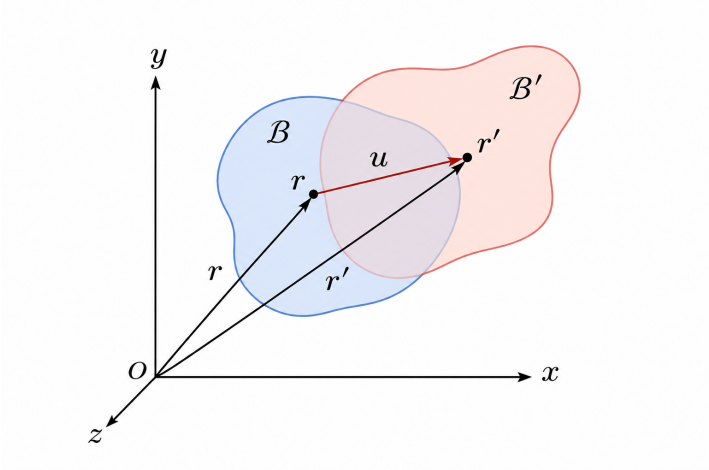


Figure 1: Deformation of a continous body

where $\mathbf{u}(\mathbf{r})$ is the displacement vector field, given by

$$\mathbf{u}(\mathbf{r}) = \begin{pmatrix} u_x(x,y,z) \\ u_y(x,y,z) \\ u_z(x,y,z) \end{pmatrix}.$$

1. Now consider an infinitesimal rectangular element of a continuous medium with dimensions dx, dy and dz.

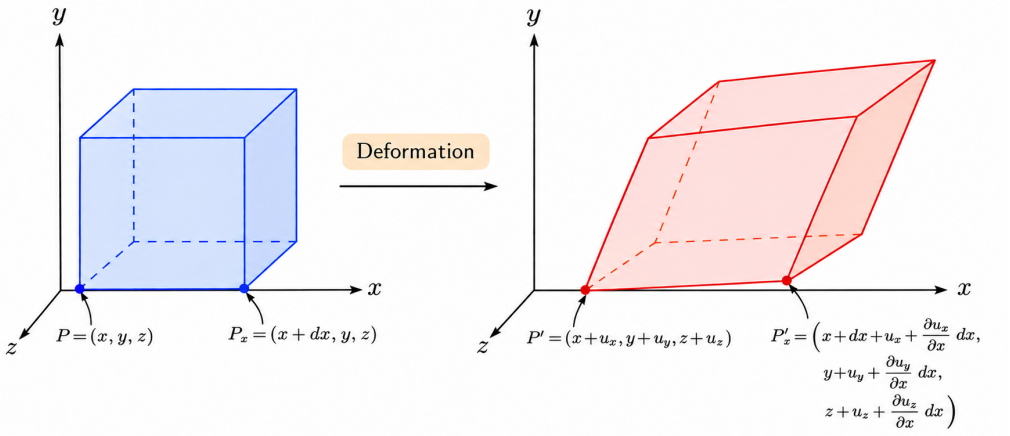


Figure 2: Deformations of a very small cuboid

Let one of its vertices be located at $P = (x, y, z)$. The three neighbouring vertices along the coordinate directions are initially located at

$$P_x = (x + dx, y, z), \qquad P_y = (x, y + dy, z), \qquad P_z = (x, y, z + dz).$$

Under a deformation, every material point is displaced according to the displacement field

$$\mathbf{u}(x,y,z) = (u_x(x,y,z),\, u_y(x,y,z),\, u_z(x,y,z))\,.$$

Consequently, the point P is mapped to

$$P' = (x + u_x(x,y,z),\, y + u_y(x,y,z),\, z + u_z(x,y,z))\,.$$

Similarly, the neighbouring point P_x is mapped to

$$P'_x = (x + dx + u_x(x + dx, y, z),\ y + u_y(x + dx, y, z),\ z + u_z(x + dx, y, z))\,.$$

Expanding the displacement components in a Taylor series about (x, y, z), and retaining only the first-order terms, we obtain the change in the position of the neighbouring point relative to P' to be:

$$P'_x - P' = \left(\left(1 + \frac{\partial u_x}{\partial x}\right) dx,\ \frac{\partial u_y}{\partial x} dx,\ \frac{\partial u_z}{\partial x} dx\right).$$

In particular, the change in the length of the edge along the x-direction, to first order, is

$$\Delta x = \frac{\partial u_x}{\partial x}\, dx.$$

2. Strain is defined as the fractional change in length, the normal strain along the x-direction is

$$\epsilon_{xx} = \frac{\Delta x}{dx} = \frac{\partial u_x}{\partial x}.$$

Following the same procedure for the edges along the y and z directions, we obtain

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}, \qquad \epsilon_{zz} = \frac{\partial u_z}{\partial z}.$$

3. From the previous calculations, we know that the change in height of the edge parallel to the x-axis is

$$\Delta y = \frac{\partial u_y}{\partial x}\, dx.$$

Therefore, for small deformations,

$$\alpha \approx \tan\alpha = \frac{\Delta y}{dx} = \frac{\partial u_y}{\partial x}.$$

Similarly, we can obtain

$$\beta \approx \tan\beta = \frac{\partial u_x}{\partial y}.$$

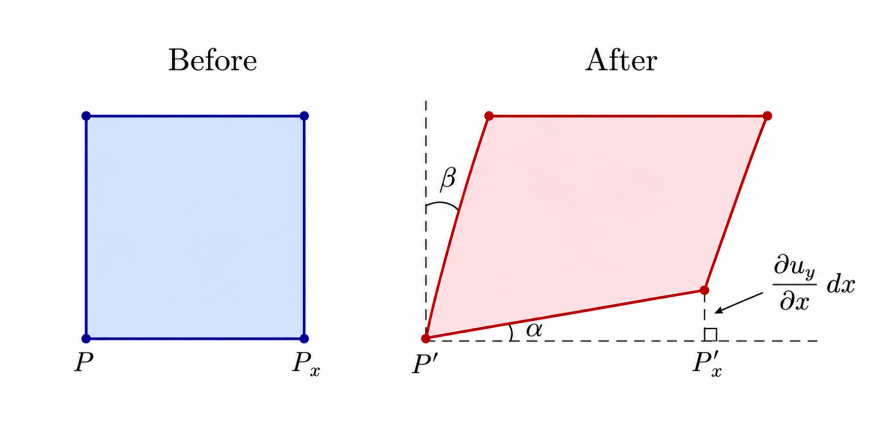


Figure 3: The shear strain

The total change in the right angle between the two edges is given by the sum of these two angles. Hence,

$$\gamma_{xy} = \alpha + \beta = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y}.$$

This quantity is called the engineering shear strain.

4. Therefore, the engineering shear strains are

$$\gamma_{xy} = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} \quad \gamma_{yx} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}.$$

By convention, the components of the infinitesimal strain tensor are defined as half the engineering shear strain. Thus,

$$\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2}\left(\frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y}\right).$$

Definition 1.1 (Strain Tensor).

The strain tensor is a second-order symmetric tensor that describes the deformation of a continuous body. It is given by

$$\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{pmatrix}.$$

The diagonal components represent normal strains, given by

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x} \quad \epsilon_{yy} = \frac{\partial u_y}{\partial y} \quad \epsilon_{zz} = \frac{\partial u_z}{\partial z}.$$

The off-diagonal components represent shear strains:

$$\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right), \quad \epsilon_{yz} = \epsilon_{zy} = \frac{1}{2}\left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y}\right), \quad \epsilon_{xz} = \epsilon_{zx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x}\right).$$

1.2 Strain

Assumption 1.2

We consider a material satisfying the following assumptions:

A1. Homogeneous. Its material properties are identical throughout.

A2. Isotropic. Its material properties are identical in every direction.

A3. Linearly Elastic. The stress is linearly related to the strain for sufficiently small deformations, allowing us to use the principle of superposition.

Young's modulus is defined as the ratio of longitudinal stress to longitudinal strain, i.e.

$$E = \frac{\text{Longitudinal Stress}}{\text{Longitudinal Strain}}.$$

When a block is stretched along a particular direction, say the x-direction, it contracts along the two perpendicular directions.

Since the material is isotropic, its elastic response is independent of the direction of loading. The Poisson's ratio (which is defined as the negative ratio of transverse strain to longitudinal strain) is thereofore constant. For a uniaxial stress in the x-direction, we have

$$\nu = -\frac{\epsilon_{yy}}{\epsilon_{xx}} \quad \nu = -\frac{\epsilon_{zz}}{\epsilon_{xx}}.$$

Similarly, for uniaxial loading along the y and z directions, the corresponding relations are

$$\nu = -\frac{\epsilon_{xx}}{\epsilon_{yy}} = -\frac{\epsilon_{zz}}{\epsilon_{yy}} \quad \nu = -\frac{\epsilon_{xx}}{\epsilon_{zz}} = -\frac{\epsilon_{yy}}{\epsilon_{zz}}.$$

1. Normal Stress Consequently, when a uniaxial stress σ is applied along the x-direction, we obtain

$$\epsilon_{xx} = \frac{\sigma}{E} \quad \epsilon_{yy} = -\frac{\nu\sigma}{E} \quad \epsilon_{zz} = -\frac{\nu\sigma}{E}.$$

Now suppose that we apply three normal stresses simultaneously, namely σ_{xx}, σ_{yy} and σ_{zz}. Since the material is linearly elastic, we can use the principle of superposition.

Consider the strain along the x-direction. The individual contributions are:

(a) The stress σ_{xx} produces a longitudinal strain

$$\epsilon_{xx}^{(1)} = \frac{\sigma_{xx}}{E}.$$

(b) The stress σ_{yy} produces a transverse strain along the x-direction

$$\epsilon_{xx}^{(2)} = -\frac{\nu\sigma_{yy}}{E}.$$

(c) Similarly, the stress σ_{zz} produces a transverse strain along the x-direction

$$\epsilon_{xx}^{(3)} = -\frac{\nu\sigma_{zz}}{E}.$$

Adding all the contributions, we obtain

$$\epsilon_{xx} = \epsilon_{xx}^{(1)} + \epsilon_{xx}^{(2)} + \epsilon_{xx}^{(3)} = \frac{1}{E}\left[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})\right].$$

By symmetry, the corresponding relations along the y and z directions are

$$\epsilon_{yy} = \frac{1}{E}\left[\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})\right], \quad \epsilon_{zz} = \frac{1}{E}\left[\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})\right].$$

2. Shear stress Hooke's law for shear is

$$\tau = G\gamma,$$

where engineering shear strain is denoted by γ, and G is the shear modulus. We also have

$$G = \frac{E}{2(1+\nu)}.$$

Theorem 1.3 (Tensor form of Hooke's law).

For an isotropic, linearly elastic body, the stress-strain relation in tensor form is given by

$$\boldsymbol{\sigma} = \lambda\ \mathrm{tr}(\boldsymbol{\epsilon})I + 2G\boldsymbol{\epsilon},$$

where

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}.$$

Here, E is Young's modulus, ν is Poisson's ratio, G is the shear modulus, and I is the identity tensor.

1.3 Newton's laws and the Navier–Cauchy equation

1. Consider a small rectangular element of a continuous body with dimensions dx, dy and dz. Its volume and mass are $dV = dx\,dy\,dz$ and $dm = \rho\,dV$, respectively.

   The element experiences two kinds of forces:

   (a) Surface forces. These are the forces exerted by the surrounding material on the faces of the element. They arise due to the stress tensor, which is given by

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}.$$

   Consider first the two faces whose normals point along the x-direction. The face at position x experiences an x-directed force

$$F_x^{(-)} = -\sigma_{xx}(x, y, z)\,dy\,dz,$$

   whereas the opposite face at $x + dx$ experiences

$$F_x^{(+)} = \sigma_{xx}(x + dx, y, z)\,dy\,dz.$$

Adding these contributions and using a first-order Taylor expansion, we obtain

$$F_x^{(1)} = \frac{\partial \sigma_{xx}}{\partial x}\, dV.$$

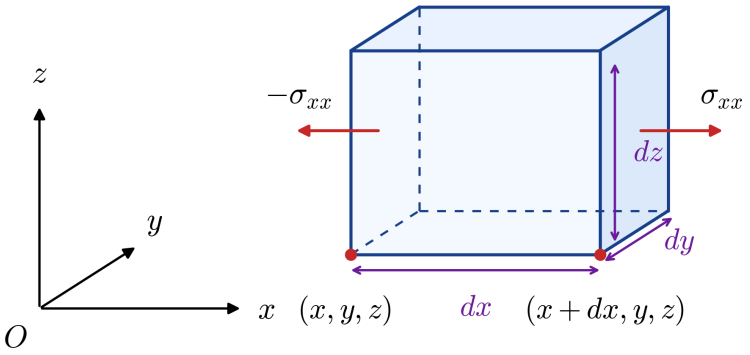


Figure 4: Force due to surrounding material

Similarly, the contributions from the two pairs of faces normal to the y- and z-directions are

$$F_x^{(2)} = \frac{\partial \sigma_{xy}}{\partial y}\, dV, \qquad F_x^{(3)} = \frac{\partial \sigma_{xz}}{\partial z}\, dV.$$

Thus, the total surface force along the x-direction is

$$F_x^{\text{surface}} = \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} \right) dV.$$

Repeating the same procedure along the y- and z-directions, we obtain the total surface force in vector form:

$$\mathbf{F}_{\text{surface}} = (\nabla \cdot \boldsymbol{\sigma})\, dV,$$

where

$$\nabla \cdot \boldsymbol{\sigma} = \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}.$$

(b) Body forces. These are forces that act throughout the volume of the element, such as gravitational forces. Let the body force per unit volume be represented by

$$\mathbf{f} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}.$$

Therefore, the total body force acting on the element is

$$\mathbf{F}_{\text{body}} = \mathbf{f}\, dV.$$

2. The net force acting on the infinitesimal element is therefore

$$\mathbf{F}_{\text{net}} = (\nabla \cdot \boldsymbol{\sigma} + \mathbf{f})\, dV.$$

Applying Newton's second law, we have

$$\rho \mathbf{a} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}.$$

Theorem 1.4 (Newton's law for a continuum).

According to Newton's second law, the equation of motion for a continuum is given by

$$\rho \mathbf{a} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f},$$

where ρ is the mass density, $\mathbf{a}$ is the acceleration vector, $\boldsymbol{\sigma}$ is the stress tensor, and $\mathbf{f}$ is the external body force per unit volume.

2 The Chladni Plate

1. Consider a thin plate of thickness h, occupying the region $-h/2 \le z \le h/2$. Initially, its middle surface lies in the xy-plane (so that any point on the middle surface is $\mathbf{r}(x, y) = (x, y, 0)$).

   Assume that after deformation, the middle surface is displaced in the z-direction by an amount $w(x, y, t)$. Its position vector therefore becomes $\mathbf{r}'(x, y, t) = (x, y, w(x, y, t))$.

   The normal vector to the deformed middle surface (assuming small slopes, that is, $|w_x|, |w_y| \ll 1$) is given by

$$\mathbf{n} = \begin{pmatrix} -w_x \\ -w_y \\ 1 \end{pmatrix}.$$

2. According to the Kirchhoff–Love hypothesis (for thin plates), any point initially lying along the normal to the middle surface moves along the normal to the deformed middle surface. Furthermore, the normal remains straight and its length remains unchanged.

   Thus, a point initially located at $\mathbf{r} = (x, y, z)$ is transported to the position

$$\mathbf{r}' = (x, y, w) + z(-w_x, -w_y, 1) = (x - zw_x,\ y - zw_y,\ w + z).$$

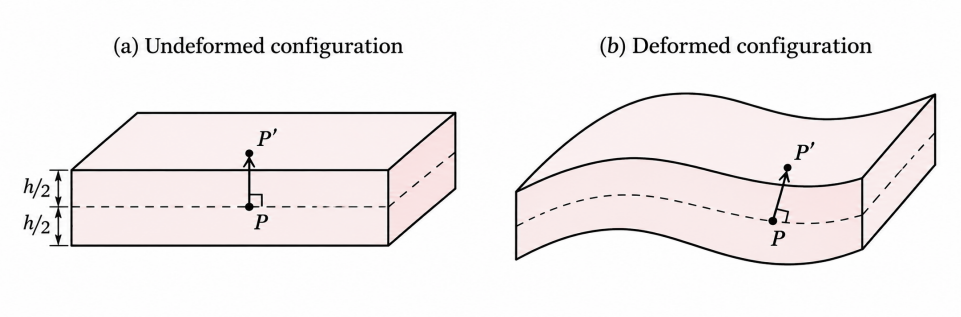


Figure 5: Illustration of the Kirchhoff-Love law (this actually is prety intuitive)

3. The displacement field is thus given by (note that $w = w(x, y, t)$):

$$\mathbf{u} = \left(-zw_x, -zw_y, w\right),$$

   The strain tensor therefore is:

$$\boldsymbol{\epsilon} = -z \begin{pmatrix} w_{xx} & w_{xy} & 0 \\ w_{xy} & w_{yy} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

4. The equations of motion for the plate are given by the three components of Newton's law:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = \rho \frac{\partial^2 u_x}{\partial t^2} = -\rho z w_{xtt},$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = \rho \frac{\partial^2 u_y}{\partial t^2} = -\rho z w_{ytt},$$

$$\frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho \frac{\partial^2 u_z}{\partial t^2} = \rho w_{tt}.$$